Rainbow perfect matchings
on even cycles and two disjoint squares

Akhil Ravuri

Abstract

We determine the maximum number of colors in an edge coloring of an even cycle that avoids a rainbow perfect matching, and count all extremal partitions of its fixed edge set. For two disjoint squares, we prove that the maximum is six and that exactly two extremal partitions occur. The arguments are elementary and self-contained; no novelty is claimed.

1 Definitions and the even-cycle formula

All graphs are finite and simple, and edge colorings need not be proper. A perfect matching is *rainbow* if its edges have pairwise distinct colors. For a graph G admitting an avoiding coloring, let $A(G)$ be the maximum number of colors in a coloring with no rainbow perfect matching. Let $N_{\mathrm{ext}}(G)$ count the color-class partitions attaining this maximum. Throughout, the edge set is fixed: color names are ignored, but partitions related by vertex symmetries are not identified.

Theorem 1. *For every integer $n \geq 2$,*

$$A(C_{2n}) = 2n - 2, \qquad N_{\mathrm{ext}}(C_{2n}) = \binom{n}{2}^2.$$

Proof. An even cycle has exactly two perfect matchings, M_1 and M_2. Once the matching edge at one vertex is chosen, the remaining choices around the cycle are forced. These are the two alternating sets of edges, and they partition $E(C_{2n})$.

Let S_i be the set of colors on M_i. In an avoiding coloring, neither matching is rainbow, so $|S_i| \leq n - 1$. Thus

$$|S_1 \cup S_2| \leq |S_1| + |S_2| \leq 2n - 2.$$

Conversely, choose one pair of edges in each matching and give that pair a common color. Give every remaining edge its own color, using disjoint palettes between the matchings. This uses $2n-2$ colors and avoids a rainbow perfect matching, proving the formula for A.

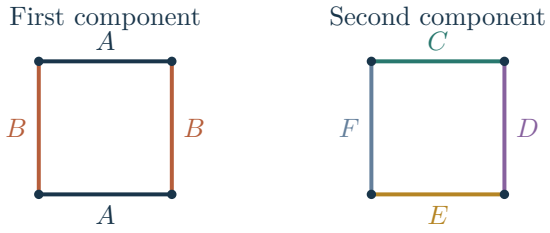
Equality forces $|S_1| = |S_2| = n-1$ and $S_1 \cap S_2 = \varnothing$. A partition of n edges into $n-1$ nonempty classes has exactly one class of size two and otherwise singletons. Hence each matching has one repeated pair. Choosing the two pairs independently gives $\binom{n}{2}^2$ distinct edge partitions, and every extremal partition arises uniquely this way. □

2 The maximum for two disjoint squares

Let $G = H_1 \sqcup H_2$, where H_1 and H_2 are fixed copies of C_4. A perfect matching of G is the union of a perfect matching in each component. There are therefore exactly four global perfect matchings.

Theorem 2. *For $G = C_4 \sqcup C_4$, we have $A(G) = 6$.*

Proof. Color the cyclically ordered edges of H_1 with A, B, A, B and give the four edges of H_2 four distinct new colors. Every global perfect matching contains one of the two monochromatic local matchings of H_1. Thus six colors are attainable without a rainbow perfect matching.



An avoiding six-coloring: the first square forces a repeated color.

For the upper bound, consider a coloring using at least seven colors on the eight edges. With eight colors, every perfect matching is rainbow. With seven, precisely two edges e, f share a color and all other classes are singletons. In the component containing e, choose the alternating matching omitting e, and combine it with either matching of the other component. The resulting perfect matching omits e, so it cannot contain the only repeated pair. It is rainbow, proving the upper bound. □

3 Characterizing the extremal partitions

Theorem 3. *With edge positions fixed and color names ignored, $N_{\text{ext}}(C_4 \sqcup C_4) = 2$.*

Proof. A six-color partition of eight edges has class sizes $s_1, \ldots, s_6$ with $\sum_{i=1}^{6}(s_i - 1) = 2$. Thus the only possible size patterns are

$$(3, 1, 1, 1, 1, 1) \quad \text{or} \quad (2, 2, 1, 1, 1, 1).$$

Excluding the triple. Every edge belongs to exactly two of the four global perfect matchings. Three equal-colored edges therefore have six total appearances across those matchings. To obstruct every matching, each would have to contain at least two of the three edges, requiring at least eight appearances. This contradiction rules out a triple.

Two repeated pairs. Let P and Q be the two doubleton color classes. A perfect matching is non-rainbow exactly when it contains P or Q in full. The number of global perfect matchings containing a given pair is as follows.

Pair position	Number of matchings
Adjacent edges in one component	0
Opposite edges in one component	2
One edge in each component	1

Adjacent edges cannot occur together. Opposite edges specify one local matching, leaving two choices in the other component. One edge in each component specifies both local choices, giving exactly one global matching.

Together, the two pairs must obstruct all four matchings. Each pair obstructs at most two, so both must obstruct exactly two, with their obstructed sets disjoint. Hence each pair consists of opposite edges in one component. If the pairs belonged to different components, choosing the other local matching in each component would give a rainbow global matching. Both pairs must therefore lie in the same component.

Since the color classes are disjoint, the pairs must be precisely that component's two alternating perfect matchings. There are two choices for this component; the other component has four singleton classes. Each choice avoids a rainbow perfect matching, and the choices define distinct fixed-edge partitions. Renaming or interchanging colors adds no further partitions, so the count is two. □

4 Small cases and computational checks

The preceding proofs give the following values.

Host G	$A(G)$	$N_{\mathrm{ext}}(G)$
C_4	2	1
C_6	4	9
C_8	6	36
C_{10}	8	100
$C_4 \sqcup C_4$	6	2

The repository's documented sparse search also checks C_4, C_6, C_8, and the disjoint squares, agreeing with these formulas. The C_{10} row is a consequence of the proof; no additional computation is claimed here. Computation is not a premise of any theorem in this note.

For the edge order, raw data, reproduction commands, and execution limits, see Discrete Math Lab, PR #1, especially the following files:

```
results/data/sparse_rainbow.json
results/sparse_verification.md
```

These counts concern color partitions on fixed edges, not graph isomorphism classes.